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Depositional Facies Controls on the Geomechanical Properties of the Middle Jurassic Tuwaiq Mountain Formation

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The geomechanical properties of rocks are crucial for understanding reservoir behavior and variations in quality, particularly in complex carbonate systems. While various approaches, such as well logs and seismic data, have been employed to characterize these properties in carbonate formations, achieving accurate results from these proxies remains challenging. The Middle Jurassic Tuwaiq Mountain Formation (TMF) in Saudi Arabia is of significant importance for both conventional and unconventional hydrocarbon exploration. However, limited research exists on the influence of depositional facies on the geomechanical properties of the TMF and the resulting implications for reservoir quality and heterogeneity. Thus, this study examines the rock mechanical behavior, geochemical composition, petrophysical properties, lithofacies and their interrelationships within the TMF. In the study area, TMF consists of three principal depositional facies, which are bioclastic Wackstone, peloidal grainstone and bioclastic-peloidal packstone. The maximum and minimum average values for uniaxial compressive strength (UCS), compressional and shear wave velocities measured were 148MPa and 34MPa; 5924m/s and 3307m/s; 3135m/s and 1861m/s, respectively. The average porosity values obtained are ranging from a minimum of 2.3% to a maximum of 19.2%. The highest value of the UCS were recorded from the bioclastic wackstone depositional facies with relatively high organic matter content and the peloidal grainstone shows the lowest UCS values. Generally, the results indicate that UCS correlates well with both depositional facies and porosity. This study provides an insight into the influence of depositional facies on the geomechanical properties of carbonate reservoirs, their implications for predicting reservoir quality and stability in the TMF and comparable analog formations. The findings contribute to a more comprehensive understanding of carbonate systems and provide valuable information for reservoir management and development strategies.

Introduction

In the oil and gas industry, the geomechanical properties of reservoirs is of paramount importance. These properties affect the structural integrity of wellbores and are essential in the development of natural fractures (Curtis, 2010). This knowledge is indispensable, especially in the context of unconventional reservoirs, where fractures act as key conduits through which oil and gas stored in the rock units flows (Abu-

Mahfouz et al., 2020). The geomechanical properties variation within formations, particularly in carbonate rocks, is crucial for various industries such as construction, mining, and petroleum (Shaban, 2021). Understanding how rocks within formations respond to different stressors is essential, as it has significant implications for various disciplines. Geologists and engineers can make informed decisions about reservoir development strategies by studying the mechanical characteristics of different rocks encountered in the field. This knowledge ultimately leads to the optimization of production and enhances overall field performance.

The TMF, which dates to Middle Jurassic period situated in central Saudi Arabia, is known for both conventional and unconventional hydrocarbon reserves (Mehrabi et al., 2014). Globally, unconventional reservoirs have demonstrated production viability through various operations, often require stimulation due to their inherently low permeability (Lee & Morse, 1999; Al-Fatlawi et al., 2019). To enhance reservoir drainage, it is common practice to implement multiple hydraulic fracturing stages. Given the complexity of carbonate systems, characterizing these reservoirs presents a significant challenge, and predicting the mechanical behavior of the reservoirs during production is another challenge. These challenges may be due to geomechanical properties variations that can occur in different depositional facies, resulting to heterogeneous deformation types and intensity due to changes in lithological, petrographic, petrophysical properties of a rock unit as it responds to production process (Zham and Hennings 2009).

Understanding the geomechanical properties of geological materials, including the various controls for their strength, is paramount from a scientific perspective. This comprehensive understanding enables better assessment of the behaviors of rocks in diverse depositional settings and effective management of different types of earth resources. Carbonate rocks are important source and reservoir rocks in several geological setting (Cole et al. 1994). The carbonate rocks usually behave differently in terms of their geomechanical behavior from other rocks, such as siliciclastic sedimentary, metamorphic or igneous rocks. These variations observe are usually connected to the biological and chemical phenomena that are associated with it, as well as the several diagenetic processes (Budai et al. 2002). The variations in rock strength can lead to a substantial impact on the mechanisms governing its behavior within different geological formations.

The UCS, along with other mechanical properties, plays a critical role in defining the strength, failure mechanisms, and fracture initiation in geological materials (Shaban, 2021). Different rock properties together control rock behavior under stress and influences fracture development. Previous studies have investigated how petrographic features relate to the geomechanical behavior of rocks, finding that mineralogical composition, grain morphology, size, and distribution significantly affect rock strength (Arif et al., 2013; Hosseini et al., 2011). Other research has examined the links between petrophysical attributes and geomechanical properties, demonstrating that as microcracks and porosity decrease, the UCS typically increase, and vice versa (Brunhoeber et al., 2020). A study by Khanlari et al. (2010) emphasized the importance of cementing materials in improving the UCS of carbonate rocks. Conversely, Ahmadi et al. (2008) observed that units of carbonate rocks with a high proportion of biogenic components often exhibit reduced strength.

This study aims to analyze the interplay between lithofacies, mechanical properties, petrophysical characteristics, and mineral composition in carbonate rocks. Therefore, we focused on an exceptionally well-exposed Middle Jurassic sequence known as the TMF, which is significant for hosting subsurface water and hydrocarbon reservoirs. The TMF has been studied for its stratigraphy, sedimentology, depositional environments, biostratigraphy, geochemistry, diagenetic features, and structural geology (Steineke, 1937; Arkell, 1952; Vaslet et al., 1983; Al-Husseini, 1997; Hughes, 2004, 2009; Helena et al., 2012; Cantrell et al., 2014; Al Ibrahim, 2014). However, there is a major gap in understanding how variations in lithofacies influence the geomechanical behavior of this formation. The findings of this research will contribute to a better understanding of how depositional facies variations influence geomechanical behavior, thereby informing predictions of reservoir quality, stability assessments, and ensuring sustainable productivity in carbonate systems.

Stratigraphy and Geological Framework

A prominent geological feature in central Saudi Arabia is the Tuwaiqescarpment, an extensive ridge extending from Al Mundfin near Sultanah and to Sharourah in the south, with its southern segment rising to an elevation of approximately 1,022 meters, located just west of the city of Riyadh (Edgell, 2006). The regional stratigraphy includes two key formations: the Wajid Sandstone and the TMF. The Wajid Sandstone Formation, which overlies the Precambrian basement rocks of the Arabian Shield, is characterized by non-marine cross-bedded sandstones that gradually transition upwards into loosely consolidated sandstone. This unit also contains common bands of conglomerates and ironstones, along with occurrences of limonitic and hematitic concretions. Its uppermost portions occasionally feature silty shale deposits (Hewaify et al., 2017). Based on its stratigraphic context, correlations with adjacent well sections, and the presence of spores, the Wajid Sandstone is generally considered to be of Lower Permian age or older (Powers et al., 1966, 1968).

In the Mid Jurassic period, deposition resulted in clastic-rich limestone formations that extended across central Arabia and adjoining regions (Powers et al., 1966). The Jurassic sequence in Saudi Arabia belongs to the Shaqra Group, including Tuwaiq Mountain (Fig. 1). The TMF unconformably overlies the Dhurma Formation (Cantrell et al., 2014). This formation is predominantly comprised of lagoonal and stromatoporoid-bearing carbonates that accumulated within a shallow-marine (Hewaify et al., 2017). Stratigraphically, the Shaqra Group is overlain by the Sulaiy Formation (Fig. 1b). Its lower boundary marks the transition between the Triassic and Jurassic periods. Each constituent formation within the Shaqra Group exhibits unique lithological frameworks and stratigraphic ages.

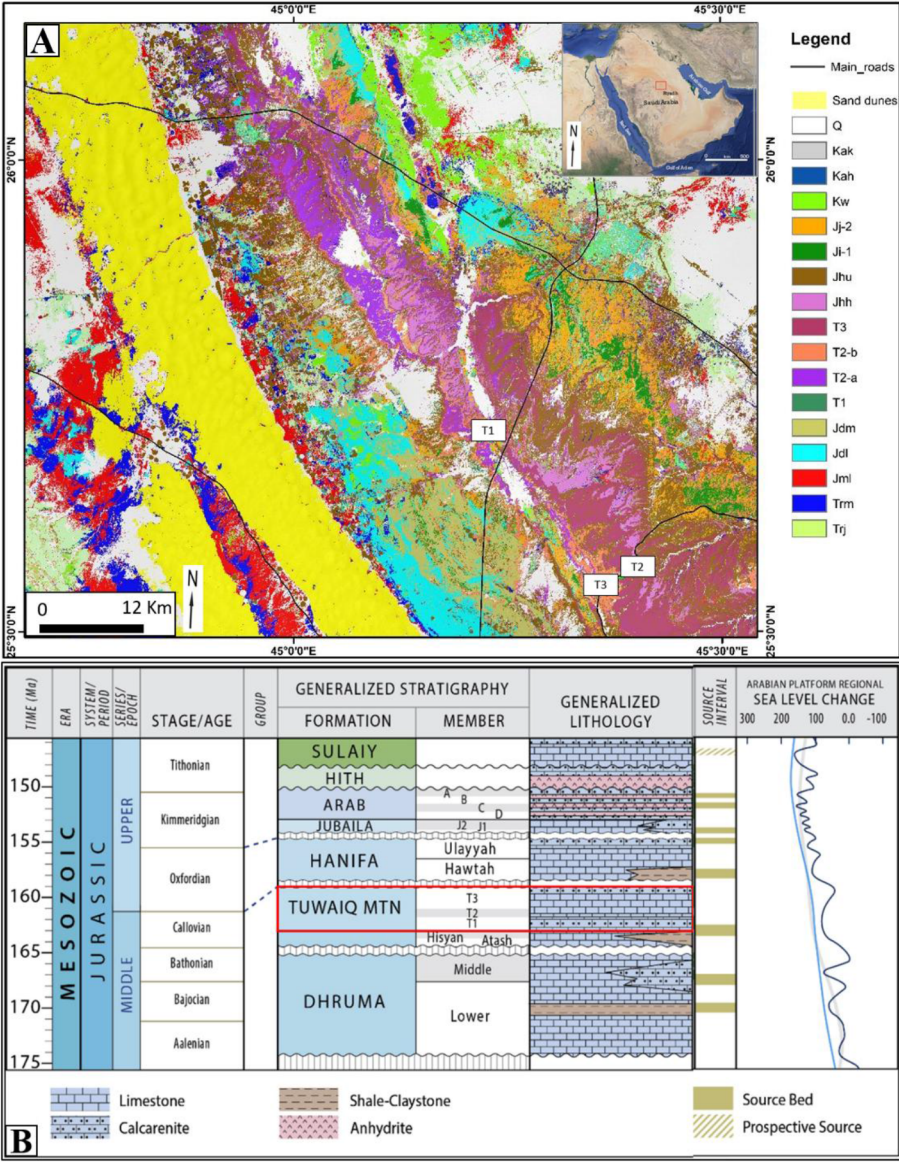


Figure 1—A) Geologic map of the Jurassic outcrops in central Saudi Arabia showing the studied outcrops, the inset map provides a regional overview. B) Stratigraphic column showing Middle and Upper Jurassic, the red boxhighlight the studied TMF (After Cantrell et al., 2014).

Materials and Methodology

This research focuses on analyzing samples collected from various sections within the TMF outcrop, located in central Saudi Arabia (Fig. 1). Representative samples spanning the TMF succession were systematically collected for detailed laboratory investigations. The analysis employed in this study include petrographic thin section analysis, compositional characterization, porosity measurements, sonic velocity testing, and UCS.

Uniaxial Compressive Strength (USC) Measurement

To quantify UCS of the studied carbonate samples, a scratching test was performed using the Epslog Wombat device. This equipment measures the intrinsic specific energy (ISE), which reflects the energy required to cut a unit volume of rock, providing a proxy for rock strength (Kolawole and Ispas, 2020). Prior to testing, ten core samples were prepared to have flat and smooth surfaces, ensuring stable placement during the

test. The ISE was calculated from the applied cutting force and geometrical parameters of the scratch, as expressed by:

$$\varepsilon = \frac{F - VI}{wd} \quad (1)$$

Here ε , represents intrinsic specific energy; F is the applied force; VI , vertical intersect; w , width of the cutter; and d indicates the depth. To minimize surface damage effects and potential cracking, approximately 2 mm of the sample surface was first removed before measurement. Multiple scratch tests with variable cutting depths (six per sample) were performed to ensure reproducibility, with the final ISE value taken as the average.

Thin Section Petrography

Thin sections, representing key facies across the formation, were prepared and examined under microscope. The petrographic study aimed to characterize the texture, microfacies, and diagenetic features of the carbonate rocks. Detailed observations included analysis of skeletal grain types, preservation states, cementation, organic matter presence, microfracture patterns, pore network distributions, and grain spatial arrangements. These petrographic data were critical for interpreting depositional environments and post-depositional processes within the TMF.

Porosity and Velocity Determination

Porosity was measured using an Automated Porosimeter-Permeameter (AP-608). Core plugs, were used for this analysis. Selected samples analyzed represent the full stratigraphic and facies range within the TMF. Ultrasonic wave velocity tests were conducted in line with International Society for Rock Mechanics (ISRM, 1981) standards. Both the compressional (V_p) and shear (V_s) waves velocities provides insights into elastic properties and rock integrity. Measurements were performed using Autolab bench-top velocity apparatus, ensuring full coupling between transducers and sample ends. Each sample was tested three times, and average velocities were reported. These data were further used to calculate Poisson's ratio computed by:

$$PR = \frac{V_p^2 - 2V_s^2}{2(V_p^2 - V_s^2)} \quad (2)$$

Young's modulus was also determined using the NER Autoscan system equipped with an impulse hammer probe, which assesses local elastic stiffness and mechanical heterogeneity. This method enables evaluation of rock strength variability at a core scale.

Mineralogical Composition Analysis

Selected samples underwent bulk mineralogical analysis using X-ray Diffraction (XRD). Powdered samples, prepared with a Retsch RM 200 mortar grinder, were analyzed using a Malvern Empyrean PANalytical diffractometer, scanning from 2° to 89.9° 2θ to capture a broad mineral range. Data interpretation was conducted with HighScore Plus (v.4.9) software against the ICDD PDF-4 2022 database.

Results

Identification and characterization of Microfacies

The rock samples were classified according to the foundational schemes by Dunham (1962) and Embry and Klován (1971). This framework facilitated a systematic identification and interpretation of microfacies, illuminating the heterogeneity within the studied formation. Three major depositional facies were distinguished including; bioclastic wackestone, peloidal grainstone, and bioclastic-peloidal packstone (Fig. 2a-c). Each microfacies exhibits unique textural and compositional attributes, revealing the complex nature

of the carbonate depositional environment. Descriptions integrate core observations and petrographic thin section analysis.

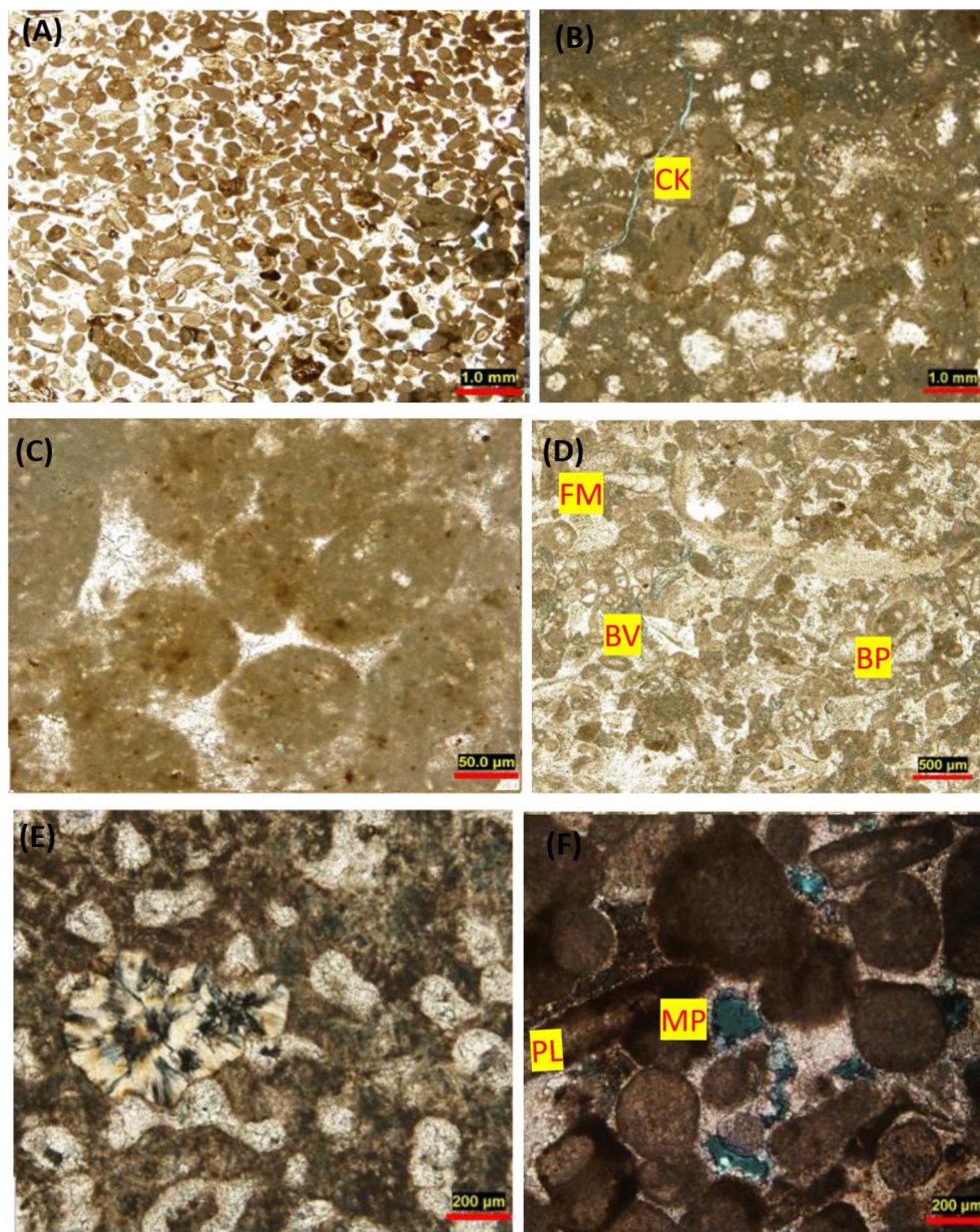


Figure 2—Thin section images of representative microfacies: (a) peloidal grainstones typical of its grainular texture and mud clasts. (b) Bioclastic wackstones with fractures (c) bioclastic-peloidal packstone (d) bioclastic wackstones with some organic matter content (e) localized silica replacement within the facies (f) pore types present in peloidal grainstone microfacies. BV; bivalves, CK; cracks and microfractures, BP; brachiopods, FM; foraminifera, PL; peloids, and MP; moldic porosity.

The Peloidal grainstone microfacies. The facies typically appears as loosely compacted, light grey to whitish rock. It is characterized by a grain-supported fabric composed predominantly of micritized peloids (100-500 µm), with occasional fragments of bivalves, brachiopods, and sparse larger mud clasts (>1 mm). Poor sorting is common, with moderate porosity evident through interparticle, vuggy, and moldic pore types. Some microporosity is present within peloids. Occasional pore-filling silica cements were observed (Fig. 2e), further modifying the pore network.

The Bioclastic-peloidal packstones. This microfacies shows a mixed texture, generally mud-supported but occasionally grain-supported in some samples. With light to dark grey colour with poor sorting.

Bioclastic debris, including brachiopod fragments, sponge spicules, gastropods, calcareous algae, and minor coral fragments, is abundant within a matrix of lime mud (Fig. 2c). Additional constituents such as micritized grains, mud clasts, and organic material are present. Porosity is predominantly intergranular and intragranular, with rare occurrences of fractures and minor vein development.

The Bioclastic wackestones microfacies. A mud-dominated matrix defines this microfacies, with both non-skeletal and skeletal grains. Skeletal components comprise coral fragments, bivalves, brachiopods (Fig. 2d). Some skeletal grains displayed micritic coatings with higher organic matter content. This facies exhibits very low porosity, mostly moldic and vuggy, primarily from the dissolution of skeletal elements. Thin sections reveal microfractures and occasional open fractures, some partially infilled with silica and organic matter. These microfractures locally enhance porosity within an otherwise tight rock matrix.

Petrophysical and Geomechanical Properties

These properties were assessed, focusing on porosity ($\varnothing$), ultrasonic velocities (compressional, V_p , and shear, V_s), and UCS measured using the Epslog device. Results are summarized in Table 1, with average UCS values presented in Fig.3. Each microfacies exhibited distinctive petrophysical and mechanical behavior.

Table 1—Summary of Average UCS, porosity, P and S wave velocities, poisons ratio and Young modulus.

Sample ID	Porosity (%)	Average UCS (Mpa)	V_p (m/s)	V_s (m/s)	Ed (GPa)	PR
Bioclastic Wackestone Facies						
S44	3.2	147	5737	2873	19.3	0.28
S77	11.8	115	5206	2603	14.07	0.3
S22	3.4	98	5924	3135	24.4	0.31
S66	2.3	132	5675	3122	45	0.26
Peloidal Grainstone Facies						
S09	19.2	41	3307	1531	14.07	0.36
S11	10.2	25	4973	2694	17.1	0.32
S33	15.8	38	4051	2008	13.5	0.34
S88	16.4	32	3903	1861	14.07	0.35
Bioclastic-peloidal Packstone Facies						
S55	5.7	71	4558	2589	197	0.29
S99	4.4	58	5671	3122	18.6	0.31

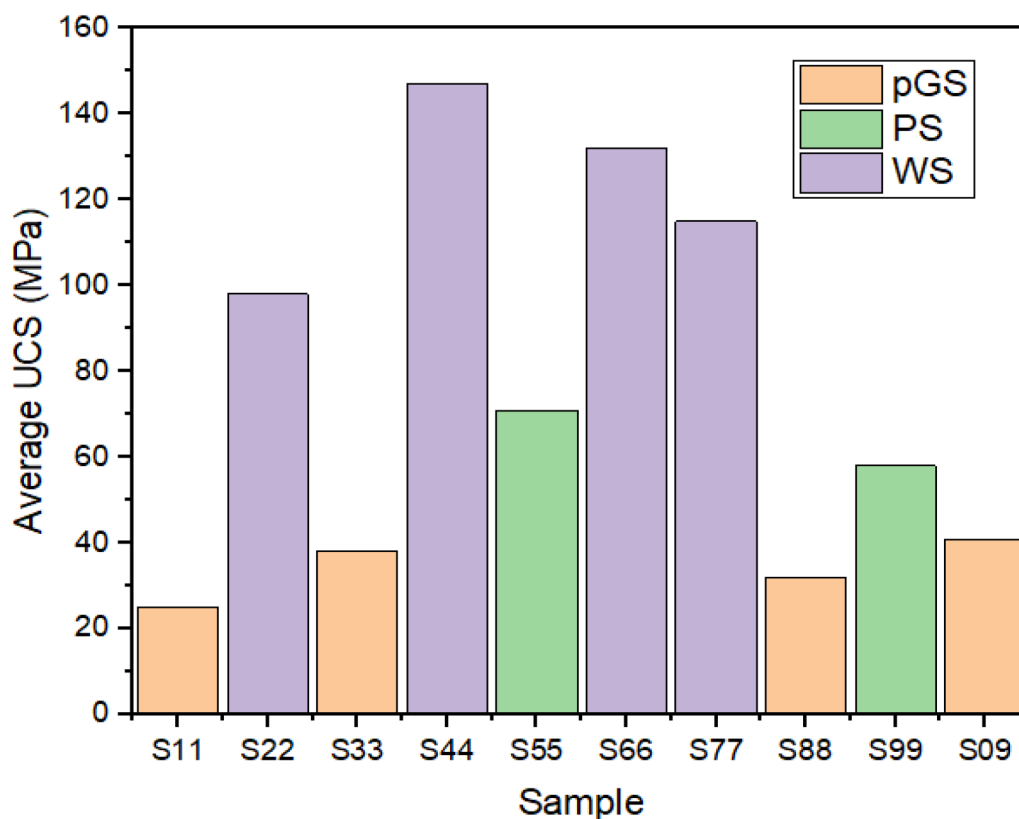


Figure 3—Bar chart for the average UCS values measured for the analyzed samples showing the different facies.

For bioclastic wackestone porosity is consistently low between 2.3% to 3.4%, except for an anomalous sample showing 11.8%. Compressional and shear wave velocities range from 5206 to 5924 m/s and 2603 to 3135 m/s, respectively. UCS values span 98 to 183.3 MPa, averaging 148 MPa. This microfacies displays the highest UCS and velocity values among the three. The porosities range from 4.4% to 5.7% were obtained for bioclastic peloidal packstone. UCS values are lower than those of bioclastic wackestone, with measurements between 21 and 87.5 MPa. V_p and V_s span 4558 to 5206 m/s and 2589 to 3122 m/s, respectively. While the peloidal grainstone facies exhibits the highest porosity of 10.2% to 19.2% and the lowest UCS values (average of 34 MPa). Compressional and shear velocities are also the lowest, with V_p ranging from 3307 to 4973 m/s; with V_s from 1531 to 2694 m/s.

Mineralogical Composition

XRD analysis indicates that calcite is the predominant mineral across all three microfacies, consistently comprising more than 90% of the total mineral content. Quartz appears in minor amounts, with the highest concentration of about 0.71% found in the bioclastic wackestone facies. QEMSCAN imagery (Fig. 4a-f) highlights the spatial variability of quartz distribution. This mineralogical data, when combined with petrographic observations, helps clarify the link between mineral composition and the geomechanical properties.

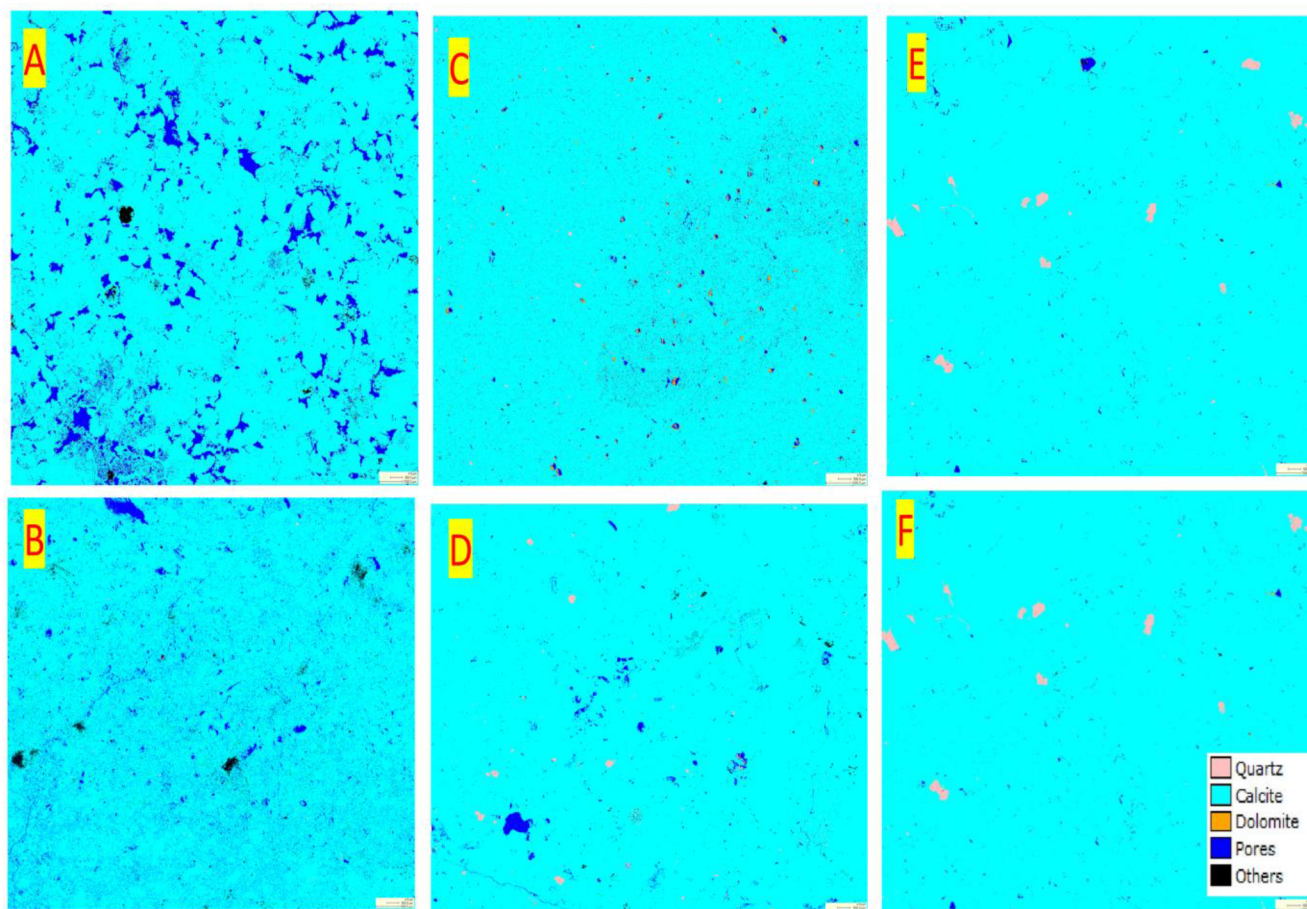


Figure 4—Mineral distribution for the three identified microfacies. (a, b) peloidal grainstone. (c, d) Bioclastic-peloidal packstone with moderate quartz mineral. (e, f) high quartz mineral content in bioclastic Wackestone.

Discussion

Geomechanical Characterization

The carbonate rocks examined in this study demonstrate clear variations in geomechanical properties that align with all depositional facies identified (Fig. 5a). Among these, bioclastic wackestone exhibits the highest mean UCS at approximately 148 MPa, followed by the bioclastic-peloidal packstone at around 64.5 MPa, and the peloidal grainstone showing the least mean UCS value of 34 MPa. According to the ISRM rock strength classification, both the bioclastic wackestone and bioclastic-peloidal packstone are categorized within the medium to high strength class and peloidal grainstone falls within the moderate strength range. The marked differences in UCS values among the facies emphasize the heterogeneity of the carbonate units, highlighting the influence of facies-related characteristics on rock mechanical behavior (Yaghoubi et al., 2023).

The differences in rock strength and stiffness among the microfacies show how their depositional textures and diagenetic changes affect their ability to sustain stress and deformation. The facies with higher UCS and Young's modulus, can better withstand stress compared to the weaker peloidal grainstone, which has higher porosity and is more likely to deform or fail under pressure. This suggests that different microfacies will behave differently during activities like hydraulic fracturing or reservoir development. Understanding these differences is important for predicting how the rock will respond in subsurface conditions and for making better decisions in resource extraction and management.

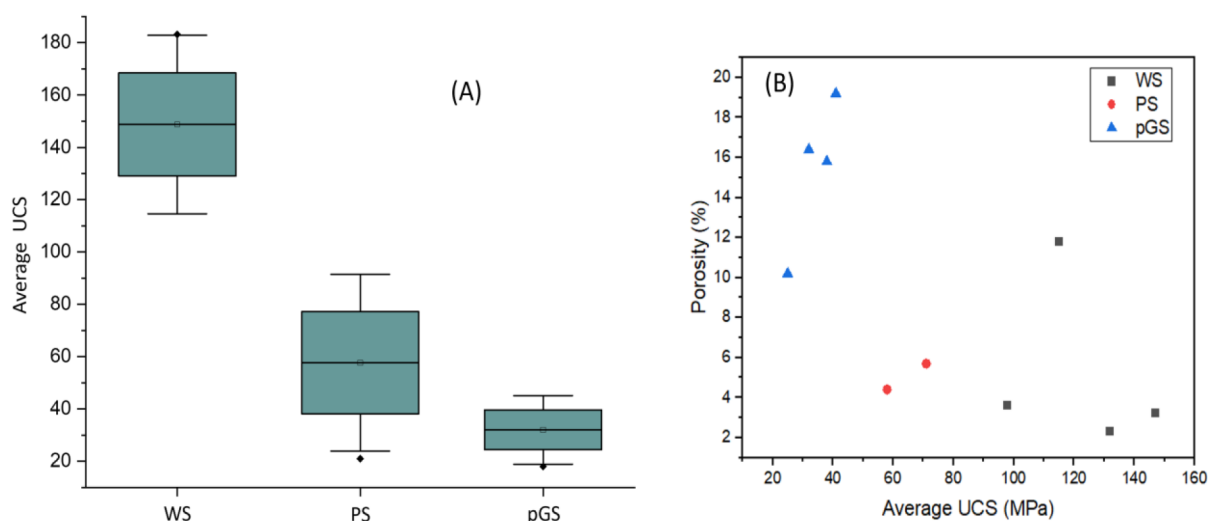


Figure 5—(A) Plots for measured UCS values categorized according to their respective facies; (B) the relationship between the porosity of each microfacies and its corresponding average UCS values.

Microfacies characteristics and geomechanical properties

Based on carbonate texture classification (Dunham, 1962; Embry and Klovan, 1971), the three depositional microfacies identified in this study show distinct geomechanical properties. The observed variation in mechanical properties (Table 1) reveals the impact of the different facies on the geomechanical properties of the studied Jurassic succession. Recent studies highlighted the influence of porosity, fabric, grain size and fractures on the strength of rocks (Abu Seif, 2016; Brunhoeber et al. 2020). In this research geomechanical properties are particularly associated to each carbonate depositional facies. The study shows that geomechanical behavior of the carbonate rocks closely relates to the depositional microfacies identified, following the classifications proposed by Dunham (1962) and Embry and Klovan (1971). Three main microfacies were distinguished, each with different mechanical properties that reflect their textures and compositions (Table 1). These results align with other research that highlights the influence of factors such as texture, porosity, grain size, and fractures on rock strength (Brunhoeber et al., 2020).

Thin-section analysis revealed that certain features within the rock can significantly affect mechanical strength. For example, the bioclastic wackestone microfacies showed medium to high UCS. This can be attributed to the presence of cementing materials that fill microfractures, which helps to bind the rock together. However, some samples also showed partially filled cracks that may lower the strength in specific cases. Khanlari et al. (2010) reported similar findings, noting that finer grain sizes and micritic cement improve rock strength. Ahmadi et al. (2008) also pointed out that the shape of fossils and their distribution can influence the mechanical properties of carbonate rocks. Our samples contained fossils which may have affected the strength observed. Different textural features among the microfacies played an important role in determining their mechanical behavior. The mud-supported bioclastic wackestones displayed the highest strength values compared to the other microfacies, likely because of their well-cemented nature. Similar observations were made by Ameen et al. (2009), who highlighted the link between rock texture and strength.

The knowledge of how microfacies relate to geomechanical properties is essential for accurately predicting rock behavior in the field. Since different facies often exhibit consistent strength values linked to their depositional features, this knowledge can be effectively used to estimate rock strength at larger scales. Such insights not only improve the characterization of mechanical behavior in carbonate reservoirs but also enhance the planning and implementation of engineering and geological projects. This understanding contributes to safer and more efficient reservoir management by enabling more targeted exploration and development strategies.

Geochemical composition and geomechanical properties

The mineralogical and geochemical composition of carbonate rocks plays a critical role in determining their geomechanical properties and overall behavior under stress. Studies has shown that minerals such as quartz and calcite exert a considerable influence on rock strength and brittleness, depending on their abundance and distribution within a given facies (Li et al., 2018). Moreover, the geochemical composition can affect porosity, which is essential in governing a rock's response to stress and deformation. Recognizing these interconnections is vital for making accurate predictions of reservoir properties and for assessing stability in relation to geomechanical behavior. UCS of the carbonate rocks was found to depend on several factors, including mineral composition, density, grain size, porosity, and microstructural characteristics (Agustawijaya, 2007). Mineralogical variations across different facies as shown by the QEMSCAN images, revealing that high strength values can be attributed to quartz content within the microfacies. A good correlation between UCS and quartz content was observed (Fig. 6a), reinforcing the significant role of quartz in enhancing rock strength. Therefore, in the peloidal grainstone, a decrease in quartz content coupled with an increase in clay minerals led to reduced UCS values, highlighting the importance of mineral composition in controlling rock mechanical properties. It is also observed that the presence of calcite laminations can reduce UCS relative to quartz-rich samples, consistent with findings by Abu Mahfouz et al. (2023). While Li and Wang (2019) suggested that calcite may not significantly affect the UCS of rocks, our results indicate that quartz is a more influential mineral in controlling the UCS of carbonate rocks. Across all examined microfacies, high calcite content did not appear to significantly impact UCS values compared to quartz.

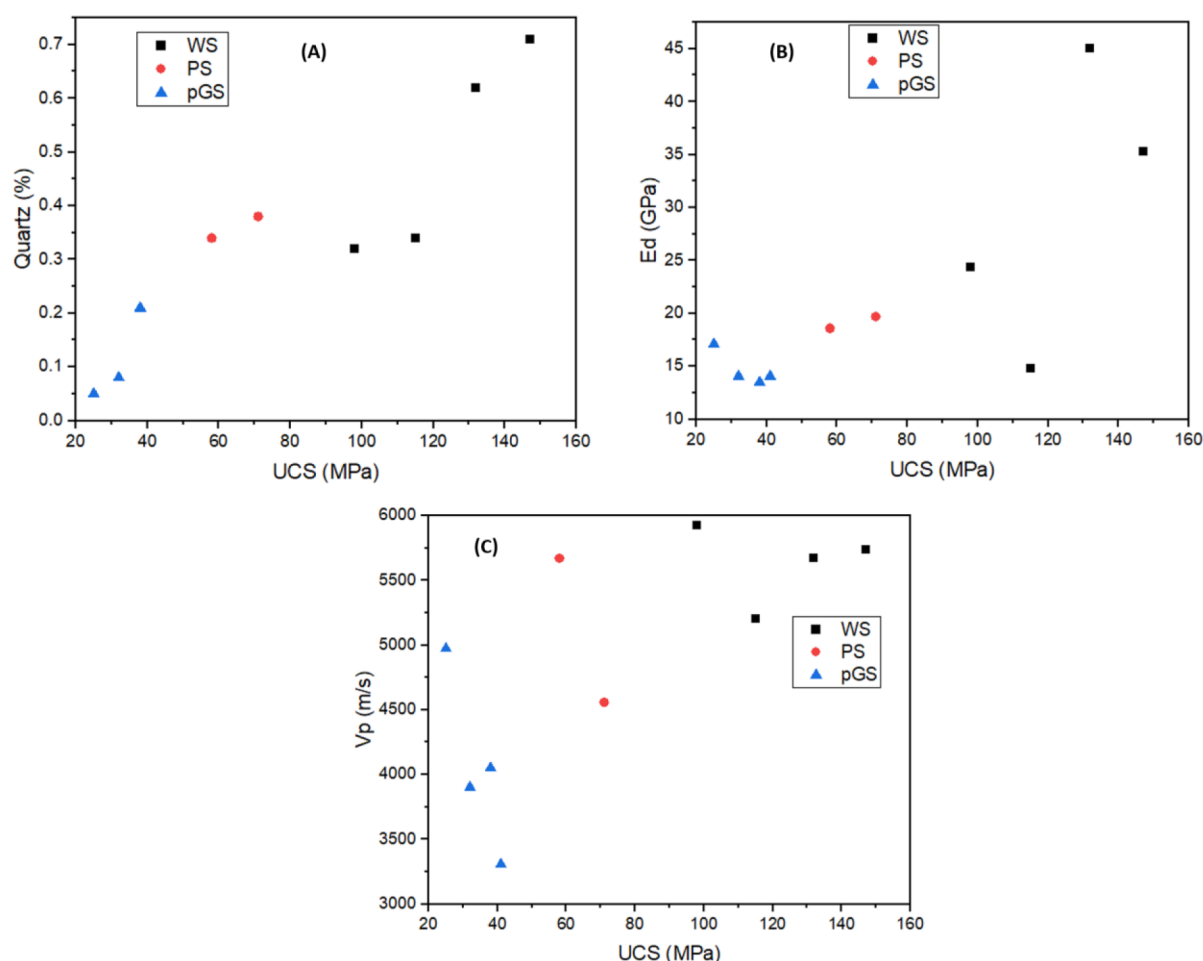


Figure 6—The graphical representations for relationship between measured UCS values: a) percentage amount of quartz mineral present, b) Young's modulus, c) compressional velocity.

The mineralogical variations within the rocks were found to affect compressional and shear-wave velocities. Quartz-rich bioclastic wackestones exhibited higher compressional-wave velocities than other facies, underscoring the contribution of quartz to both rock stiffness and wave propagation. In contrast, calcite, though abundant in all the facies, did not show a noticeable impact on either compressional or shear-wave velocities. These observations align with the findings of [Murphy et al. \(2013\)](#) on carbonate rocks. Young's modulus, which reflects the stiffness of rocks, is influenced by mineral type, density, cementation, matrix structure, cracks, and porosity ([Agustawijaya, 2007](#)). In carbonate rocks, mineralogical composition has a major impact on Young's modulus, which in turn affects seismic wave velocities. These findings are consistent with the work of [Dong et al. \(2017\)](#), who emphasized that quartz typically contributes to higher bonding strength and rock stiffness compared to other minerals like clay, this demonstrates that mineralogical and geochemical factors can influence the geomechanical behaviour of rocks behavior in different depositional environments and conditions.

Rock Mechanical Properties and Petrophysical Parameters

The relationship between geomechanical behavior and petrophysical parameters is valuable for evaluating the mechanical response of rocks in different microfacies. Petrophysical parameters are closely linked to how rocks respond to external stresses, making their assessment essential for accurate predictions of reservoir performance, stability analyses([Ganat, 2020](#)).

The present study identifies a systematic correlation between porosity and UCS across the studied carbonate facies ([Fig. 7a](#)). An increase in porosity from 2.3% to 19.2% corresponds with a decline in UCS values in the microfacies. This trend reflects the progressive weakening of rocks from the dense bioclastic wackestone facies to the more porous peloidal grainstone facies. A similar decrease is observed in Young's modulus, confirming the inverse relationship between porosity and rock stiffness. This finding aligns with previous studies emphasizing porosity as a dominant factor in controlling geomechanical strength ([Wang, 2019](#)). While this relationship is consistent, some variability is observed in the data, which may be attributed to differences in pore types, including moldic and fracture porosity identified in the microfacies. These pore structures exert a considerable influence on rock mechanical response by modifying how stress is distributed and accommodated within the rock matrix, thereby impacting deformation and compaction processes ([Bashah and Pierson, 2012](#)). An inverse relationship between sonic wave velocities and porosity is also evident from the data ([Fig. 8a and b](#)), with both compressional and shear velocities decreasing as porosity increases. This observation is consistent with the findings of [Ameen et al. \(2009\)](#) and reinforces the significance of porosity in modulating wave propagation characteristics. Moreover, the heterogeneity of porosity distribution within facies such as the bioclastic wackestone reflects the complex interplay of depositional and diagenetic processes that shape carbonate rock properties. This complexity contributes to the observed heterogeneity in the geomechanical behavior of the TMF carbonates.

The bioclastic wackestone microfacies, characterized by low porosity and a high degree of compaction, exhibit higher Young's modulus values compared to the more porous peloidal grainstones. This finding suggests that low-porosity facies possess greater stiffness and are more resistant to deformation. The association between lower UCS values and higher Poisson's ratio in the peloidal grainstone samples ([Fig. 8c](#)) further demonstrates the impact of porosity on mechanical properties. The analyzed samples having porosity value below 5.5% fall into the medium-high strength category, providing a useful threshold for assessing strength in similar carbonate systems. It is important to recognize that while porosity exerts a strong influence on rock stiffness and strength, the relationship is not strictly linear. Additional factors including mineral types, pore type, and rock texture also contribute significantly to strength variations and heterogeneity within carbonate formations. Furthermore, fracture porosity can create planes of weakness that may interconnect under stress, linking previously isolated microfractures and thereby enhancing fluid flow and potentially improving reservoir quality ([Yaghoubi et al., 2023](#)). These results demonstrate that porosity,

when considered in conjunction with other petrophysical factors, plays a central role in determining the mechanical behavior of carbonate rocks.

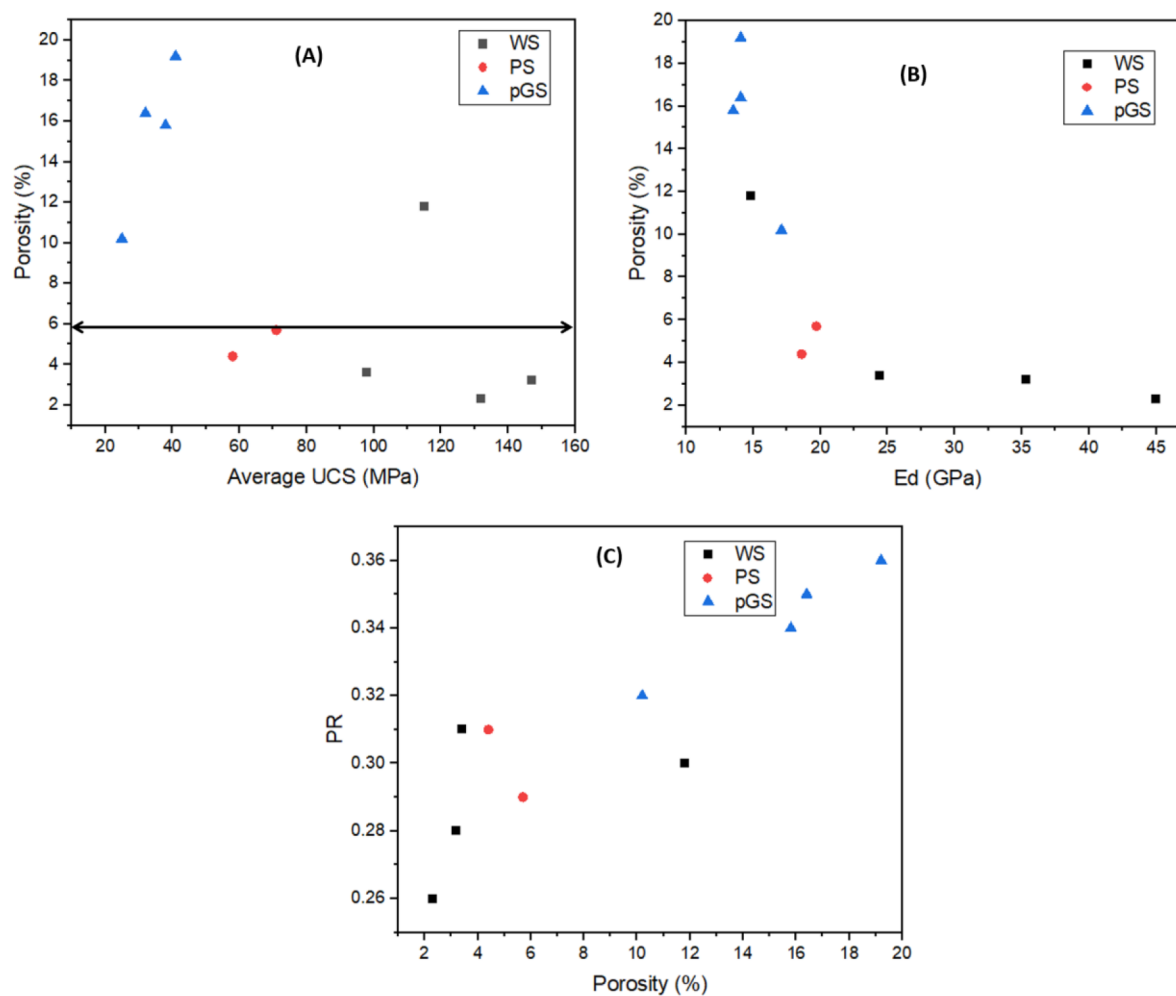


Figure 7—Chart illustrating the relationship between porosity values and: A) UCS, B) vs Young's modulus C) vs poisson ratio of the rock samples. (arrow line is the rocks sample with porosity threshold of medium rock strength)

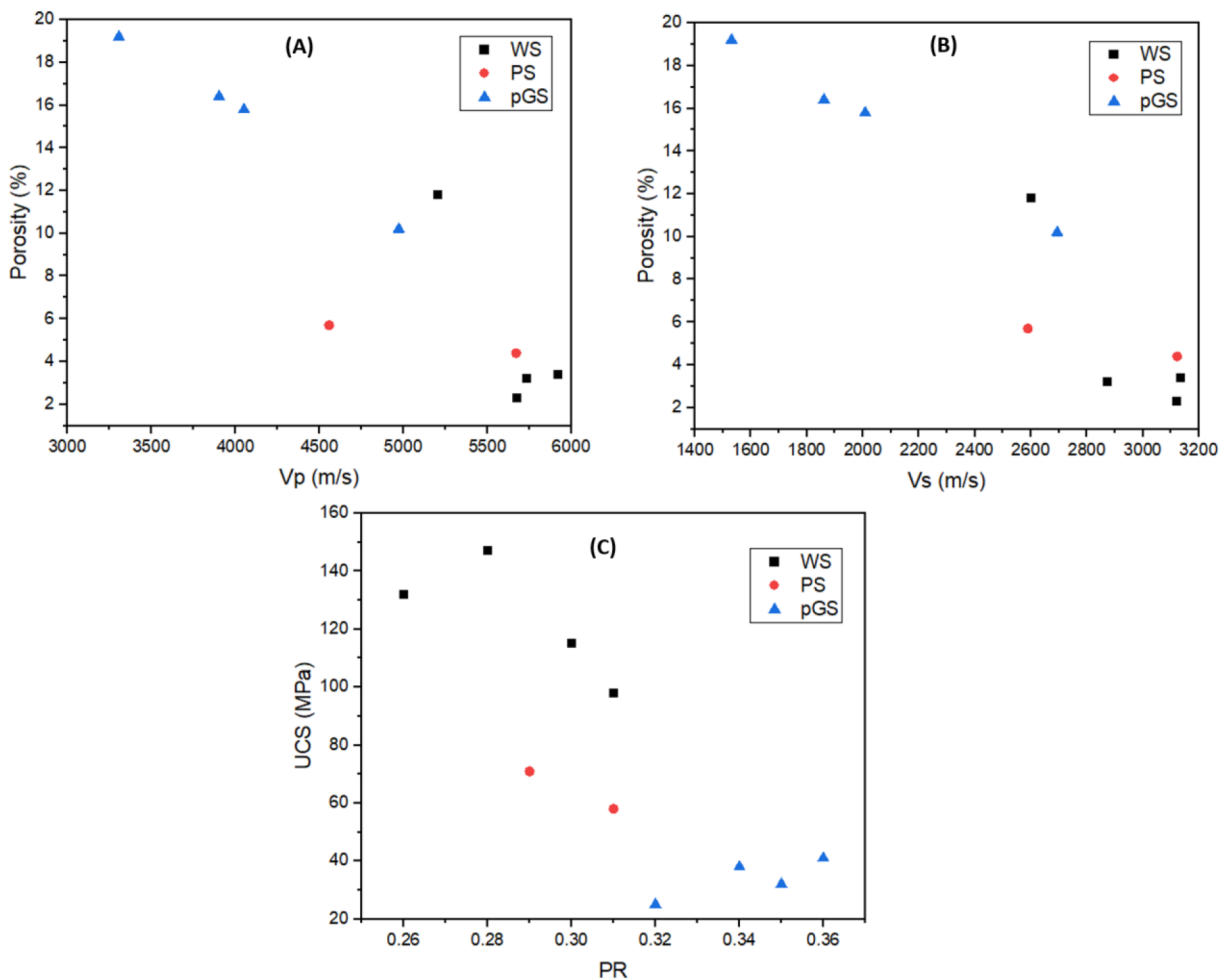


Figure 8—Graphs illustrating the relationship between porosity: a) compressional wave velocity (b) shear wave velocity and (c) UCS and Poisson ratio.

Conclusion

This study demonstrates that the Middle Jurassic TMF in central Saudi Arabia comprises three major depositional facies, each exhibiting variable geomechanical properties. The UCS ranges from 34 MPa in the peloidal grainstone to 148 MPa in the bioclastic wackestone, reflecting the heterogeneity inherent to carbonate formations. Quartz content, although minor compared to dominant calcite, emerges as a significant contributor to rock strength. Petrophysical parameters, especially porosity, exhibit a strong but non-linear influence on mechanical behavior, shaped by mineralogy, rock texture, and pore type. Compact, low-porosity facies show higher Young's modulus and lower Poisson's ratio, reinforcing the importance of integrating facies analysis with geochemical and petrophysical evaluations. This holistic approach provides a more reliable basis for geomechanical modeling and reservoir management in complex carbonate systems.

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